

Laboratorium 9 — bagging, lasy losowe i MARS (zadania)

Statystyka w SI i analizie danych — rozszerzenie laboratorium 8

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Contents

Przygotowanie — <code>CarseatsH</code> i drzewa (lab 8)	2
Przygotowanie — <code>Boston</code> , drzewa i <code>lm</code> (lab 8)	3
Zadanie 1 — <code>Boston</code> : rozszerzenie porównania MSE	4
Zadanie 2 — <code>CarseatsH</code> : rozszerzenie porównania błędu klasyfikacji	5
Zadanie 3 — Istotność predyktorów: <code>lm</code> / <code>glm</code> vs <code>rpart</code> vs las vs MARS	6
Zadanie 4 — <code>Hitters</code> i <code>Titanic</code> (bagging, las, MARS, MARS d2)	10
Hitters — MSE na zbiorze testowym	10
Titanic — błąd na holdoutcie z <code>titanic_train</code> + predykcje na <code>titanic_test</code>	11
Podsumowanie	12

Pakiety: `rpart`, `rpart.plot`, `ISLR`, `MASS`, `randomForest`, `earth`, `plotmo`, `titanic`.

```
library(rpart)
library(rpart.plot)
library(ISLR)
library(MASS)
library(randomForest)
library(earth)
library(plotmo)
library(titanic)
```

Zadania wg materiału *Bagging, lasy losowe i MARS — zadania*:

1. Rozszerzyć zadania 1, 2 i 3 z laboratorium 8 o **bagging** (`randomForest` z `mtry` równym liczbie predyktorów), **las losowy** (domyślne `mtry`) oraz **MARS z poziomem interakcji 2** (`earth(..., degree = 2)`); dla pełnego obrazu porównawczego uwzględniam także **MARS addytywny** (domyślny `degree`).
2. Wykonać zadanie 4 z laboratorium 8 (**Hitters**, **Titanic**) stosując bagging, las losowy, MARS addytywny i MARS z `degree = 2`.

Podziały `train/test` jak w `lab8_drzewa_decyzyjne.Rmd`: `Boston` — `set.seed(2025)`; `CarseatsH` — `set.seed(1)`; `Hitters` — `set.seed(3030)`; holdout `Titanica` z `titanic_train` — `set.seed(4040)`.

Przygotowanie — CarseatsH i drzewa (lab 8)

```
CarseatsH <-  
  Carseats |>  
  transform(High = factor(ifelse(Sales <= 8, "No", "Yes")))  
  
set.seed(1)  
n_cs <- nrow(CarseatsH)  
train_cs <- sample(n_cs, 0.8 * n_cs)  
test_cs <- setdiff(seq_len(n_cs), train_cs)  
  
sales_high_big <- rpart(High ~ . - Sales, data = CarseatsH, subset = train_cs)  
sales_high_big_predicted <- predict(sales_high_big, newdata = CarseatsH[test_cs, ], type = "class")  
err_big_cs <- mean(sales_high_big_predicted != CarseatsH$High[test_cs])  
  
n_min_cs <- which.min(sales_high_big$cptable[, "xerror"])  
cp_min_cs <- sales_high_big$cptable[n_min_cs, "CP"]  
sales_high_pruned_min <- prune(sales_high_big, cp = cp_min_cs)  
sales_high_pruned_min_predicted <- predict(  
  sales_high_pruned_min,  
  newdata = CarseatsH[test_cs, ],  
  type = "class"  
)  
err_pruned_min_cs <- mean(sales_high_pruned_min_predicted != CarseatsH$High[test_cs])  
  
limit_cs <- sum(sales_high_big$cptable[n_min_cs, c("xerror", "xstd")])  
n_1se_cs <- min(which(sales_high_big$cptable[, "xerror"] < limit_cs))  
cp_1se_cs <- sales_high_big$cptable[n_1se_cs, "CP"]  
sales_high_pruned_1se <- prune(sales_high_big, cp = cp_1se_cs)  
sales_high_pruned_1se_predicted <- predict(  
  sales_high_pruned_1se,  
  newdata = CarseatsH[test_cs, ],  
  type = "class"  
)  
err_pruned_1se_cs <- mean(sales_high_pruned_1se_predicted != CarseatsH$High[test_cs])  
  
glm_high <- glm(High ~ . - Sales, family = binomial, data = CarseatsH, subset = train_cs)  
phat <- predict(glm_high, newdata = CarseatsH[test_cs, ], type = "response")  
glm_pred <- ifelse(phat > 0.5, "Yes", "No")  
glm_pred <- factor(glm_pred, levels = c("No", "Yes"))  
err_glm_cs <- mean(glm_pred != CarseatsH$High[test_cs])  
  
lda_high <- lda(High ~ . - Sales, data = CarseatsH, subset = train_cs)  
lda_pred <- predict(lda_high, CarseatsH[test_cs, ])$class  
err_lda_cs <- mean(lda_pred != CarseatsH$High[test_cs])  
  
qda_high <- qda(High ~ . - Sales, data = CarseatsH, subset = train_cs)  
qda_pred <- predict(qda_high, CarseatsH[test_cs, ])$class  
err_qda_cs <- mean(qda_pred != CarseatsH$High[test_cs])  
  
p_cs <- ncol(CarseatsH) - 2L  
  
rf_cs <- randomForest(High ~ . - Sales, data = CarseatsH, subset = train_cs, importance = TRUE)  
pred_rf_cs <- predict(rf_cs, newdata = CarseatsH[test_cs, ])
```

```

err_rf_cs <- mean(pred_rf_cs != CarseatsH$High[test_cs])

bag_cs <- randomForest(High ~ . - Sales, data = CarseatsH, subset = train_cs, mtry = p_cs)
pred_bag_cs <- predict(bag_cs, newdata = CarseatsH[test_cs, ])
err_bag_cs <- mean(pred_bag_cs != CarseatsH$High[test_cs])

mars_cs <- earth(High ~ . - Sales, data = CarseatsH, subset = train_cs, glm = list(family = binomial))
pr_mars_cs <- predict(mars_cs, newdata = CarseatsH[test_cs, ], type = "response")
pred_mars_cs <- factor(ifelse(pr_mars_cs > 0.5, "Yes", "No"), levels = c("No", "Yes"))
err_mars_cs <- mean(pred_mars_cs != CarseatsH$High[test_cs])

mars2_cs <- earth(
  High ~ . - Sales,
  data = CarseatsH,
  subset = train_cs,
  degree = 2,
  glm = list(family = binomial)
)
pr_mars2_cs <- predict(mars2_cs, newdata = CarseatsH[test_cs, ], type = "response")
pred_mars2_cs <- factor(ifelse(pr_mars2_cs > 0.5, "Yes", "No"), levels = c("No", "Yes"))
err_mars2_cs <- mean(pred_mars2_cs != CarseatsH$High[test_cs])

```

Przygotowanie — Boston, drzewa i lm (lab 8)

```

set.seed(2025)
n_bo <- nrow(Boston)
train_bo <- sample(n_bo, 0.8 * n_bo)
test_bo <- setdiff(seq_len(n_bo), train_bo)

medv_big <- rpart(medv ~ ., data = Boston, subset = train_bo)
medv_big_pred <- predict(medv_big, newdata = Boston[test_bo, ])
mse_big_bo <- mean((medv_big_pred - Boston$medv[test_bo])^2)

n_min_bo <- which.min(medv_big$cptable[, "xerror"])
limit_bo <- sum(medv_big$cptable[n_min_bo, c("xerror", "xstd")])
n_1se_bo <- min(which(medv_big$cptable[, "xerror"] < limit_bo))
cp_1se_bo <- medv_big$cptable[n_1se_bo - 1, "CP"]
medv_pruned <- prune(medv_big, cp = cp_1se_bo)
medv_pruned_pred <- predict(medv_pruned, newdata = Boston[test_bo, ])
mse_pruned_bo <- mean((medv_pruned_pred - Boston$medv[test_bo])^2)

lm_medv <- lm(medv ~ ., data = Boston, subset = train_bo)
lm_pred <- predict(lm_medv, newdata = Boston[test_bo, ])
mse_lm_bo <- mean((lm_pred - Boston$medv[test_bo])^2)

p_bo <- ncol(Boston) - 1L

rf_bo <- randomForest(medv ~ ., data = Boston, subset = train_bo, importance = TRUE)
pred_rf_bo <- predict(rf_bo, newdata = Boston[test_bo, ])
mse_rf_bo <- mean((pred_rf_bo - Boston$medv[test_bo])^2)

```

```

bag_bo <- randomForest(medv ~ ., data = Boston, subset = train_bo, mtry = p_bo)
pred_bag_bo <- predict(bag_bo, newdata = Boston[test_bo, ])
mse_bag_bo <- mean((pred_bag_bo - Boston$medv[test_bo])^2)

mars_bo <- earth(medv ~ ., data = Boston, subset = train_bo)
pred_mars_bo <- predict(mars_bo, newdata = Boston[test_bo, ])
mse_mars_bo <- mean((pred_mars_bo - Boston$medv[test_bo])^2)

mars2_bo <- earth(medv ~ ., data = Boston, subset = train_bo, degree = 2)
pred_mars2_bo <- predict(mars2_bo, newdata = Boston[test_bo, ])
mse_mars2_bo <- mean((pred_mars2_bo - Boston$medv[test_bo])^2)

```

Zadanie 1 — Boston: rozszerzenie porównania MSE

```

tab_zad1 <- data.frame(
  model = c(
    "drzewo pelne",
    "drzewo przyciete (1 SE)",
    "regresja liniowa",
    "bagging (randomForest, mtry = p)",
    "las losowy (randomForest)",
    "MARS addytywny",
    "MARS degree = 2"
  ),
  MSE_test = c(
    mse_big_bo,
    mse_pruned_bo,
    mse_lm_bo,
    mse_bag_bo,
    mse_rf_bo,
    mse_mars_bo,
    mse_mars2_bo
  )
)
knitr::kable(tab_zad1, digits = 4, caption = "Boston: MSE na zbiorze testowym (lab 8 + bagging / RF / MARS)")

```

Table 1: Boston: MSE na zbiorze testowym (lab 8 + bagging / RF / MARS)

model	MSE_test
drzewo pelne	24.0698
drzewo przyciete (1 SE)	27.0883
regresja liniowa	23.9903
bagging (randomForest, mtry = p)	10.9446
las losowy (randomForest)	11.0566
MARS addytywny	16.2006
MARS degree = 2	15.9526

Wniosek: Las i bagging często poprawiają lub dorównują pojedynczemu drzewu; MARS (degree = 2)

bywa silniejszy na nieliniowościach, ale na konkretnym podziale warto czytać liczby z tabeli.

Zadanie 2 — CarseatsH: rozszerzenie porównania błędu klasyfikacji

```
tab_zad2 <- data.frame(  
  model = c(  
    "drzewo pelne (T0)",  
    "drzewo cp_min",  
    "drzewo 1SE",  
    "glm",  
    "lda",  
    "qda",  
    "bagging (randomForest, mtry = p)",  
    "las losowy (randomForest)",  
    "MARS addytywny (GLM)",  
    "MARS degree = 2 (GLM)"  
  ),  
  blad_testowy = c(  
    err_big_cs,  
    err_pruned_min_cs,  
    err_pruned_1se_cs,  
    err_glm_cs,  
    err_lda_cs,  
    err_qda_cs,  
    err_bag_cs,  
    err_rf_cs,  
    err_mars_cs,  
    err_mars2_cs  
  )  
)  
knitr::kable(tab_zad2, digits = 4, caption = "CarseatsH: blad klasyfikacji na zbiorze testowym")
```

Table 2: CarseatsH: blad klasyfikacji na zbiorze testowym

model	blad_testowy
drzewo pelne (T0)	0.2500
drzewo cp_min	0.2625
drzewo 1SE	0.2625
glm	0.1375
lda	0.1000
qda	0.1500
bagging (randomForest, mtry = p)	0.1875
las losowy (randomForest)	0.2000
MARS addytywny (GLM)	0.1250
MARS degree = 2 (GLM)	0.1375

Wniosek: Ensemble (bagging, RF) często stabilizuje predykcję względem pojedynczego drzewa; MARS z GLM łączy hinge functions z logit — `degree = 2` pozwala na interakcje na skali liniowej predykcji MARS.

Zadanie 3 — Istotność predyktorów: lm / glm vs rpart vs las vs MARS

Jak w lab 8: testy w modelach liniowych mierzą wpływ przy ustalonej postaci parametrycznej; `variable.importance` w `rpart` oraz `%IncMSE` / `IncNodePurity` w `lesie` i `evimp` w `MARS` odzwierciedlają inne mechanizmy selekcji i interakcji.

```
slm <- summary(lm_medv)
pvals <- coef(slm)[, "Pr(>|t|)"]
imp_bo <- medv_big$variable.importance
imp_bo_norm <- 100 * imp_bo / sum(imp_bo)

imp_rf <- importance(rf_bo, type = 1)[, 1]
imp_rf_norm <- 100 * imp_rf / sum(imp_rf)

ei_bo <- evimp(mars2_bo)
ei_bo_mat <- as.matrix(ei_bo)
ei_bo_tab <- data.frame(
  zmienna = rownames(ei_bo_mat),
  as.data.frame.matrix(ei_bo_mat, stringsAsFactors = FALSE),
  check.names = FALSE
)

knitr::kable(
  data.frame(
    zmienna = names(pvals)[-1],
    p_wartosc = unname(pvals[-1]),
    istotna_05 = unname(pvals[-1]) < 0.05
  ),
  digits = 4,
  caption = "Boston: istotnosc w lm (bez intercept)"
)
```

Table 3: Boston: istotnosc w lm (bez intercept)

zmienna	p_wartosc	istotna_05
crim	0.0506	FALSE
zn	0.0140	TRUE
indus	0.7279	FALSE
chas	0.0012	TRUE
nox	0.0000	TRUE
rm	0.0000	TRUE
age	0.7930	FALSE
dis	0.0000	TRUE
rad	0.0002	TRUE
tax	0.0274	TRUE
ptratio	0.0000	TRUE
black	0.0038	TRUE
lstat	0.0000	TRUE

```
knitr::kable(
  data.frame(
    zmienna = names(imp_bo_norm),
```

```

waznosc_rpart_proc = round(imp_bo_norm, 2)
),
caption = "Boston: variable.importance drzewa (pelne na zbiorze uczacych), % sumy"
)

```

Table 4: Boston: variable.importance drzewa (pelne na zbiorze uczacych), % sumy

	zmienna	waznosc_rpart_proc
rm	rm	31.59
lstat	lstat	23.90
indus	indus	7.91
nox	nox	6.86
dis	dis	6.59
age	age	5.72
tax	tax	5.63
ptratio	ptratio	5.19
crim	crim	3.43
zn	zn	1.95
rad	rad	0.75
chas	chas	0.48

```

knitr::kable(
  data.frame(
    zmienna = names(imp_rf_norm),
    pctIncMSE_proc = round(imp_rf_norm, 2)
  ),
  caption = "Boston: %IncMSE lasu (randomForest, zbior uczacy), % sumy"
)

```

Table 5: Boston: %IncMSE lasu (randomForest, zbior uczacy), % sumy

	zmienna	pctIncMSE_proc
crim	crim	7.62
zn	zn	1.47
indus	indus	5.83
chas	chas	2.51
nox	nox	8.76
rm	rm	17.34
age	age	6.67
dis	dis	9.59
rad	rad	3.98
tax	tax	6.49
ptratio	ptratio	7.57
black	black	5.13
lstat	lstat	17.04

```

knitr::kable(
  ei_bo_tab,
  digits = 4,

```

```
caption = "Boston: evimp dla MARS (degree = 2, zbior uczacy)"
)
```

Table 6: Boston: evimp dla MARS (degree = 2, zbior uczacy)

	zmienna	col	used	nsubsets	gcv	gcv.match	rss	rss.match
rm	rm	6	1	18	100.0000	1	100.0000	1
lstat	lstat	13	1	17	64.7783	1	65.2990	1
tax	tax	10	1	15	38.7644	1	40.0219	1
nox	nox	5	1	14	31.8263	1	33.4017	1
dis	dis	8	1	13	29.9968	1	31.4159	1
age	age	7	1	11	25.3550	1	26.6544	1
rad	rad	9	1	9	20.6943	1	21.9834	1
ptratio	ptratio	11	1	6	12.5804	1	14.3248	1

```
sglm <- summary(glm_high)
p_glm <- coef(sglm)[, "Pr(>|z|)"]
imp_cs <- sales_high_big$variable.importance
imp_cs_norm <- 100 * imp_cs / sum(imp_cs)

imp_rf_cs <- importance(rf_cs, type = 1)[, 1]
imp_rf_cs_norm <- 100 * imp_rf_cs / sum(imp_rf_cs)

ei_cs <- evimp(mars2_cs)
ei_cs_mat <- as.matrix(ei_cs)
ei_cs_tab <- data.frame(
  zmienna = rownames(ei_cs_mat),
  as.data.frame.matrix(ei_cs_mat, stringsAsFactors = FALSE),
  check.names = FALSE
)

knitr::kable(
  data.frame(
    zmienna = rownames(coef(sglm))[-1],
    p_wartosc = as.numeric(p_glm[-1]),
    istotna_05 = as.numeric(p_glm[-1]) < 0.05
  ),
  digits = 4,
  caption = "Carseats: istotnosc w glm (bez intercept)"
)
```

Table 7: Carseats: istotnosc w glm (bez intercept)

zmienna	p_wartosc	istotna_05
CompPrice	0.0000	TRUE
Income	0.0000	TRUE
Advertising	0.0000	TRUE
Population	0.5851	FALSE
Price	0.0000	TRUE
ShelveLocGood	0.0000	TRUE
ShelveLocMedium	0.0000	TRUE

zmienna	p_wartosc	istotna_05
Age	0.0000	TRUE
Education	0.9224	FALSE
UrbanYes	0.4852	FALSE
USYes	0.2859	FALSE

```
knitr::kable(
  data.frame(
    zmienna = names(imp_cs_norm),
    waznosc_rpart_proc = round(imp_cs_norm, 2)
  ),
  caption = "Carseats: variable.importance drzewa (pelne), % sumy"
)
```

Table 8: Carseats: variable.importance drzewa (pelne), % sumy

	zmienna	waznosc_rpart_proc
Price	Price	32.87
ShelveLoc	ShelveLoc	21.36
CompPrice	CompPrice	12.38
Advertising	Advertising	8.48
Age	Age	8.06
US	US	5.47
Income	Income	4.20
Education	Education	3.52
Population	Population	3.19
Urban	Urban	0.46

```
knitr::kable(
  data.frame(
    zmienna = names(imp_rf_cs_norm),
    MeanDecreaseAccuracy_proc = round(imp_rf_cs_norm, 2)
  ),
  caption = "Carseats: MeanDecreaseAccuracy lasu, % sumy"
)
```

Table 9: Carseats: MeanDecreaseAccuracy lasu, % sumy

	zmienna	MeanDecreaseAccuracy_proc
CompPrice	CompPrice	9.92
Income	Income	5.37
Advertising	Advertising	12.60
Population	Population	-3.88
Price	Price	32.25
ShelveLoc	ShelveLoc	31.36
Age	Age	8.77
Education	Education	0.58
Urban	Urban	-0.88
US	US	3.92

```
knitr::kable(
  ei_cs_tab,
  digits = 4,
  caption = "Carseats: evimp dla MARS (degree = 2, GLM, zbior uczacy)"
)
```

Table 10: Carseats: evimp dla MARS (degree = 2, GLM, zbior uczacy)

	zmienna	col	used	nsubsets	gcv	gcv.match	rss	rss.match
ShelveLocGood	ShelveLocGood	6	1	12	100.0000	1	100.0000	1
Price	Price	5	1	11	86.2659	1	86.9245	1
CompPrice	CompPrice	1	1	10	75.6340	1	76.7474	1
Advertising	Advertising	3	1	9	65.4390	1	67.1395	1
Age	Age	8	1	8	53.2037	1	56.1610	1
ShelveLocMedium	ShelveLocMedium	7	1	7	46.0145	1	49.3415	1
Income	Income	2	1	5	28.8124	1	34.1114	1

Wniosek: Zmienne o wysokim %IncMSE mogą różnić się od istotnych w `lm/glm` (inna struktura, progi, interakcje); `evimp` dla MARS z GLM pokazuje kolejność włączania hinge w kontekście logitu.

Zadanie 4 — Hitters i Titanic (bagging, las, MARS, MARS d2)

Hitters — MSE na zbiorze testowym

```
set.seed(3030)
Hitters_clean <- na.omit(Hitters)
n_h <- nrow(Hitters_clean)
train_h <- sample(n_h, 0.8 * n_h)
test_h <- setdiff(seq_len(n_h), train_h)

p_h <- as.integer(length(labels(terms(Salary ~ ., data = Hitters_clean))))

fit_bag_h <- randomForest(Salary ~ ., data = Hitters_clean, subset = train_h, mtry = p_h)
fit_rf_h <- randomForest(Salary ~ ., data = Hitters_clean, subset = train_h)
fit_mars_h <- earth(Salary ~ ., data = Hitters_clean, subset = train_h)
fit_mars2_h <- earth(Salary ~ ., data = Hitters_clean, subset = train_h, degree = 2)

mse_bag_h <- mean((predict(fit_bag_h, Hitters_clean[test_h, ]) - Hitters_clean$Salary[test_h])^2)
mse_rf_h <- mean((predict(fit_rf_h, Hitters_clean[test_h, ]) - Hitters_clean$Salary[test_h])^2)
mse_mars_h <- mean((predict(fit_mars_h, Hitters_clean[test_h, ]) - Hitters_clean$Salary[test_h])^2)
mse_mars2_h <- mean((predict(fit_mars2_h, Hitters_clean[test_h, ]) - Hitters_clean$Salary[test_h])^2)

tab_hit <- data.frame(
  model = c("bagging", "las losowy", "MARS addytywny", "MARS degree = 2"),
  MSE_test = c(mse_bag_h, mse_rf_h, mse_mars_h, mse_mars2_h)
)
knitr::kable(tab_hit, digits = 4, caption = "Hitters: MSE na holdoucie (80/20, set.seed(3030))")
```

Table 11: Hitters: MSE na holdoucie (80/20, set.seed(3030))

model	MSE_test
bagging	139537.7
las losowy	135628.9
MARS addytywny	231750.0
MARS degree = 2	343967.6

Titanic — błąd na holdoucie z titanic_train + predykcje na titanic_test

Zbiór `titanic_test` z pakietu CRAN **nie** zawiera `Survived`; ocenę klasyfikacji liczę na podzbiorze testowym z `titanic_train` (jak w lab 8). Na `titanic_test` pokazuję wyłącznie przykładowe etykiety modelu.

```
vars_ti <- c("Survived", "Pclass", "Sex", "Age", "SibSp", "Parch", "Fare", "Embarked")
ti <- na.omit(titanic_train[, vars_ti])
ti$Survived_f <- factor(ifelse(ti$Survived == 1L, "Yes", "No"))
ti$Sex <- factor(ti$Sex)
ti$Embarked <- factor(ti$Embarked)
ti$Pclass <- factor(ti$Pclass)
```

```
set.seed(4040)
n_ti <- nrow(ti)
train_i <- sample(n_ti, 0.8 * n_ti)
test_i <- setdiff(seq_len(n_ti), train_i)
```

```
p_ti <- length(attr(terms(Survived_f ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked, data = ti)
```

```
ti_bag <- randomForest(Survived_f ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked, data = ti, su
ti_rf <- randomForest(Survived_f ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked, data = ti, sub
ti_mars <- earth(
```

```
  Survived_f ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked,
  data = ti,
  subset = train_i,
  glm = list(family = binomial)
)
```

```
ti_mars2 <- earth(
  Survived_f ~ Pclass + Sex + Age + SibSp + Parch + Fare + Embarked,
  data = ti,
  subset = train_i,
  degree = 2,
  glm = list(family = binomial)
)
```

```
pred_bag_ti <- predict(ti_bag, ti[test_i, ], type = "class")
pred_rf_ti <- predict(ti_rf, ti[test_i, ], type = "class")
pr_m <- predict(ti_mars, ti[test_i, ], type = "response")
pr_m2 <- predict(ti_mars2, ti[test_i, ], type = "response")
pred_mars_ti <- factor(ifelse(pr_m > 0.5, "Yes", "No"), levels = c("No", "Yes"))
pred_mars2_ti <- factor(ifelse(pr_m2 > 0.5, "Yes", "No"), levels = c("No", "Yes"))
```

```
err_bag_ti <- mean(pred_bag_ti != ti$Survived_f[test_i])
err_rf_ti <- mean(pred_rf_ti != ti$Survived_f[test_i])
```

```
err_mars_ti <- mean(pred_mars_ti != ti$Survived_f[test_i])
err_mars2_ti <- mean(pred_mars2_ti != ti$Survived_f[test_i])

tab_ti <- data.frame(
  model = c("bagging", "las losowy", "MARS addytywny (GLM)", "MARS degree = 2 (GLM)",
    blad_testowy = c(err_bag_ti, err_rf_ti, err_mars_ti, err_mars2_ti)
)
knitr::kable(tab_ti, digits = 4, caption = "Titanic (fragment titanic_train): blad klasyfikacji na holdoutcie")
```

Table 12: Titanic (fragment titanic_train): blad klasyfikacji na holdoutcie

model	blad_testowy
bagging	0.2378
las losowy	0.2098
MARS addytywny (GLM)	0.2168
MARS degree = 2 (GLM)	0.1958

```
vars_te <- c("Pclass", "Sex", "Age", "SibSp", "Parch", "Fare", "Embarked")
te_raw <- titanic_test[, vars_te]
te_raw$Sex <- factor(te_raw$Sex, levels = levels(ti$Sex))
te_raw$Embarked <- factor(te_raw$Embarked, levels = levels(ti$Embarked))
te_raw$Pclass <- factor(te_raw$Pclass, levels = levels(ti$Pclass))

pred_te <- predict(ti_rf, newdata = te_raw, type = "class")
head(data.frame(PassengerId = titanic_test$PassengerId, pred_Survived_rf = pred_te), 12)
```

```
## PassengerId pred_Survived_rf
## 1 892 No
## 2 893 No
## 3 894 No
## 4 895 No
## 5 896 No
## 6 897 No
## 7 898 No
## 8 899 No
## 9 900 Yes
## 10 901 No
## 11 902 <NA>
## 12 903 No
```

Wniosek: Dla Hitters cztery podejścia nonparametryczne / semiparametryczne porównuje się MSE na jednym podziale. Dla Titanica pełna ocena na `titanic_test` nie jest możliwa bez etykiet; raportowany błąd dotyczy holdoutu z `titanic_train`.

Podsumowanie

Rozszerzono zadania 1–3 z laboratorium 8 o bagging, las losowy oraz MARS (addytywny i `degree = 2`) na ustalonych podziałach danych; w zadaniu 3 dodano wagność z `randomForest` i `evimp` z `earth`. Zadanie 4 powtórzono dla Hitters i Titanica z czterema wariantami modeli zgodnie z poleceniem.